

Prime Palindrome Demonstration

Title: Prime Palindrome Checker using Scanner[cite: 15]

Objective/Aim:

To write a Java program to check whether a given number is both prime and palindrome[cite: 15].

System Requirements :

Hardware

- Standard PC/Laptop[cite: 1]
- Intel Core i3/i5 processor or equivalent[cite: 1, 15]
- Minimum 4 GB / 8 GB RAM[cite: 1, 15]
- Minimum 500 MB free storage[cite: 1]
- Keyboard and Monitor[cite: 1]

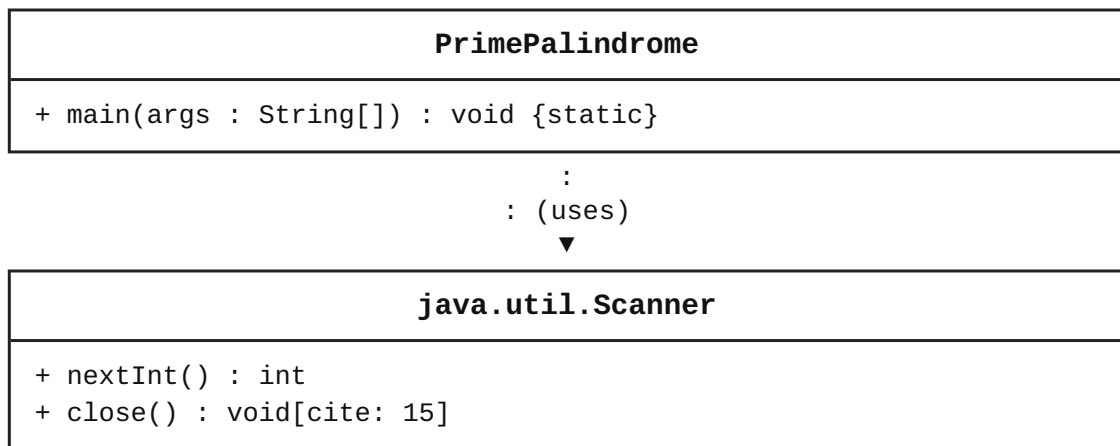
Software

- Operating System: Windows/Linux/macOS[cite: 1, 15]
- Java Development Kit (JDK 17 or JDK 21)[cite: 1, 15]
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code[cite: 1, 15]
- Command Prompt/Terminal[cite: 1]

Algorithm / Logic Description:

1. Import the `Scanner` class to read the number from the user[cite: 15].
2. Create a `Scanner` object and read the input number[cite: 15].
3. Initialize `COUNT` to zero and check all numbers from 1 to the given number[cite: 15].
4. Increment `COUNT` whenever the number is exactly divisible by the loop variable[cite: 15].
5. Reverse the given number using a `while` loop[cite: 15].
6. Compare the divisor count with 2 and compare the original number with its reverse[cite: 15].
7. If both conditions are true, display **Prime Palindrome**; otherwise, display **Not a Prime Palindrome**[cite: 15].
8. Close the `Scanner` object[cite: 15].
9. Stop the program[cite: 1].

UML class Diagram



Relationship: The `PrimePalindrome` class depends on and uses the `java.util.Scanner` class to take user input from the console[cite: 15].

Source Code:

```
import java.util.Scanner;

public class PrimePalindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int count = 0;

        for (int i = 1; i <= num; i++) {
            if (num % i == 0)
                count++;
        }

        int temp = num;
        int rev = 0;

        while (temp > 0) {
            int rem = temp % 10;
            rev = rev * 10 + rem;
            temp = temp / 10;
        }

        if (count == 2 && num == rev)
            System.out.println("Prime Palindrome");
        else
            System.out.println("Not a Prime Palindrome");
    }
}
```

```
        sc.close();  
    }  
}
```

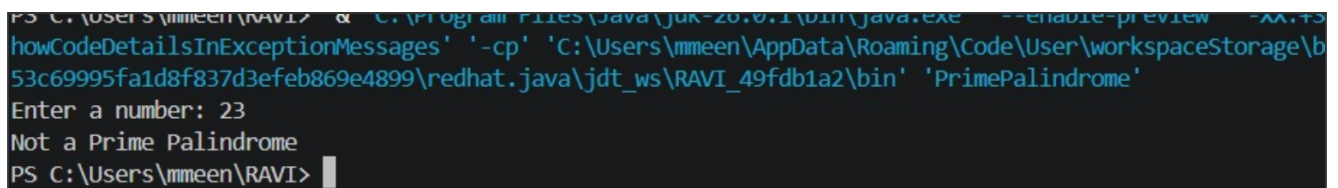
Compilation & Execution Commands:

Save file as: PrimePalindrome.java[cite: 16]

Compilation: javac PrimePalindrome.java[cite: 16]

Execution: java PrimePalindrome[cite: 16]

Output Screenshot:



```
PS C:\Users\mmmeen\RAVI> & "C:\Program Files\Java\jdk-20.0.1\bin\java.exe" --enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\mmmeen\AppData\Roaming\Code\User\workspaceStorage\b53c69995fa1d8f837d3efeb869e4899\redhat.java\jdt_ws\RAVI_49fdb1a2\bin" "PrimePalindrome"  
Enter a number: 23  
Not a Prime Palindrome  
PS C:\Users\mmmeen\RAVI>
```

Result:

The Prime Palindrome program was successfully implemented in Java. The program demonstrates reading user input using the Scanner class, checking prime divisibility using iteration, verifying palindrome equality through numerical reversal, and outputting the corresponding status[cite: 15, 16].